JAI PARASHAR

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RESEARCH INTERESTS

Theory–experiment interface of condensed matter and quantum physics, with a focus on quantum materials and devices. Interested in research that combines analytical and numerical modeling with hands-on instrumentation and hardware validation, and that could lead to new devices and technologies.

EDUCATION

BS-MS (Integrated), Physics *2023 – Present*

Indian Institute of Science Education and Research (IISER) Tirupati, Andhra Pradesh

- CGPA: 7.7/10 (through Semester 6)

Higher Secondary Education (CBSE) *2022*

Kendriya Vidyalaya No. 3, Gwalior, Madhya Pradesh

- *Class XII: 92.17%*

RESEARCH EXPERIENCE

Summer Research Intern — Quantum Transport *May – Jul 2025*

Indian Institute of Technology (BHU) Varanasi, Department of Physics — Supervisor: Dr. Panch Ram

- Computed Andreev reflection, normal reflection, and electron and hole transmission probabilities across the superconducting gap using the analytical BTK expressions in Python (transparent interface, Z = 0).
- Solved the Bogoliubov–de Gennes equations numerically (SymPy, NumPy) by matching wavefunctions at the interface; the results reproduced the BTK energy dependence, with probability conservation checked.
- Wrote a 21-page report covering the theory, the code, and a comparison of the two methods.

Summer Research Intern — Radio Astronomy Instrumentation *May – Jul 2026*

Tata Institute of Fundamental Research (TIFR), Mumbai — Supervisor: Dr. Bhaswati Mookerjea

- Studied the theory of digital spectrometers (Nyquist sampling, FFT, polyphase filterbanks, vector accumulation) and its use in the RFSoC-based backend of the BVEX balloon-borne VLBI telescope.
- Built a polyphase-filterbank + FFT spectrometer (RFDC → Munge → PFB → FFT → Power → Vector Accumulator → BRAM) on an AMD-Xilinx RFSoC4x2 board with the CASPER/Simulink toolflow, adapted from the CASPER tutorial design.
- Wrote Python (casperfpga) scripts to generate a 393.216 MHz test tone and read out the spectrum; a DAC–ADC loopback recovered it as the dominant peak, and the DAC output measured about 393.95 MHz on an oscilloscope.
- Debugged RFDC configuration problems (tile channels, port connections, Xilinx- vs. Simulink-type signals), attributed spurs at 245.76 MHz spacing to board clock leakage, and documented the fixes.

TECHNICAL SKILLS

- **Programming:** Python (NumPy, SciPy, SymPy, Matplotlib, Pandas)
- **Instrumentation & Hardware:** FPGA design via the CASPER/Simulink toolflow, AMD-Xilinx Vivado, casperfpga (Python), RFSoC4x2 board bring-up, oscilloscopes and multimeters, breadboarding and circuit debugging
- **Scientific Tools:** LaTeX, Mathematica, MATLAB/Simulink
- **Experimental Skills:** Signal measurement, error estimation, data analysis and visualization

RELEVANT COURSEWORK

- **Physics:** Quantum Mechanics I & II, Statistical Mechanics I & II, Electrodynamics I, Optics, Condensed Matter Physics I, Atomic and Molecular Physics, Computational Methods, Machine Learning, Photonics and Nonlinear Optics.
- **Mathematics & Chemistry:** Calculus, Linear Algebra, Probability & Statistics, Real Analysis, Mathematical Methods; Surface and Solid State Chemistry.
- **Labs:** Experimental physics labs (mechanics, optics, E&M, modern physics/electronics), oscilloscope measurements, report writing.

CERTIFICATIONS & LANGUAGES

- Rocket Propulsion — ISRO Certified
- Languages: Hindi (native), English (advanced), German (A1, Goethe Institut)

ACTIVITIES, OUTREACH & INTERESTS

- Volunteer, UNMOCHAN: taught science and mathematics to underprivileged children.
- Volunteer, IISER Tirupati Science Day 2025: demonstrated chemiluminescence and the Hall effect, and explained the Hall effect's history and real-world applications.
- Volunteer, NSS cleaning drive.
- Coordinator, Vivante 2026 (annual cultural fest of IISER Tirupati, 22–25 Oct 2026): built the official website (vivante26.in) using AI-assisted development tools and created social media content.
- Outside hitter in volleyball and 100 m / 200 m sprinter for 3 years; represented IISER Tirupati at IISM 2025 (bronze in athletics); multiple gold/bronze medals at regional KVS sports meets (2017–2020).
- Interests: Yoga, Weight training, Photography (landscapes and portraits), Acting, short-form content creation.